

MSCI 242M 2023 Coursework: Health Survey Analysis

The Health Survey for England (HSE) is a series of annual surveys about the health of people living in England. The survey is sponsored by the Department of Health (DoH) to provide better and more reliable information about various aspects of people's health and to monitor selected health targets, such as drinking, smoking and weight.



Department
of Health

A supplied datafile by the DoH provides 25 key pieces of health data, for over 5,000 individuals, including information on weight and BMI, plus activity habits relating to drinking, smoking and exercise.



Part A

[40 marks]

Counts

1. How many people are listed in the dataset?
2. How many males are listed in the dataset?
3. How many females are listed in the dataset?
4. How many people aged under 18 are listed in the dataset?
5. How many people are married?
6. How many people aged under 18 are married?
7. How many households are there?
8. How many 1-person households are there?
9. How many 1-person households contain a person who is single?
10. How many single people live in a 2-person household?

Averages¹ - see footnote 1 below

11. What is the average age of the respondents? Round the answer to 2 decimal places²
12. What is the average number of people per household? Round to 3 decimal places²
13. What is the average BMI of married people? Round down to the nearest integer²
14. What is the average BMI of married females? Round down to the nearest integer²
15. What is the median units of alcohol per week for all people 18 years or older?

Extremes

16. What is the highest observed alcohol units?
17. What is the pserial for the person with the highest observed alcohol units³?
18. What is the weight of the heaviest female, in kg?
19. What is the age of the oldest female?
20. How big is the household (size) where the oldest male⁴ lives?

¹ Omit any '**non-valid**' data from your average calculations, due to a response of 'Not applicable', 'Don't know' etc.

² Using an Excel function. **Do not** simply *format* the cell to appear rounded, as this will receive reduced marks.

³ If more than one person shares this value, then present the first value that appears in the datablock

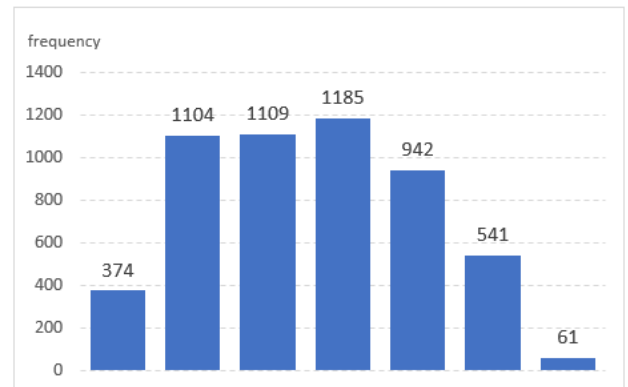
⁴ If more than one male shares this age, then present the largest household size for all these males



Part B

[20 marks]

21. What is the age of the youngest current smoker?
22. What bin width, in years, would produce this histogram for age?
23. What is the household size for the tallest female³ in the survey?
24. What is the pserial for the youngest divorced male³?
25. How many people provided a 'valid' response to all 21 survey questions from Q4 to Q25?



Note: 'Don't know', 'Not applicable', 'Not sure' and 'Refuse to answer', or a blank cell, are all examples of a non-valid response.



Part C

[20 marks]

26. Clearly, some strategic health authorities have more respondents than others. Limiting your analysis to health authorities with at least 200 respondents in the survey, which authority (by name) has the 'worst' self-assessed general health¹, on average(Q13)?
27. An alternative method of assessing poor health is to look at the number of respondents who rate their own general health as 'bad' or 'very bad' (Q13). Again, looking only at health authorities with at least 200 respondents in the survey, which authority (by name) has the highest percentage of their respondents¹ rating their health as 'bad' or 'very bad'?
28. How many all-female 3-person households are there?
29. How many people live in all-female households?
30. What is the age of the oldest person in the largest all-female household⁵?



Part D

[20 marks]

This section is concerned with alcohol consumption for women aged 40 and above.

Create a macro to produce a new datablock on a new sheet named 'Part D data' that contains **only** the following data: [10 marks]

- hserial
- sex (exclude all males)
- age (exclude any people under 18 years old)
- drating (exclude any people who responded with a -1)

How many households in the new datablock contain females aged 40 years or above with a drating above 21 units per week⁶? [10 marks]

⁵ If there are multiple households that meet this criteria, find the oldest person across all these households

⁶ This represents a recommended 'safe' limit of alcohol consumption per week



Data and Groups

Group size for this coursework can be any size between **1 and 4** members.

Once you have formed a group, submit the names of the people in your group to the 242 moodle – see the post titled ‘Coursework Group Choice’.

Each group is identified by a **group number**, for example group 56.

The **dataset** you must use is the same as your **group number**, for example 56.xlsx.

If you have any questions regarding this then you must contact Adam *before* submission.

All datasets are different, so using the incorrect dataset will generate incorrect results.

All datasets are held on moodle in a folder called ‘Coursework Data’



Support

Clarification of the tasks, data or wording of the questions can be done in the **workshop** sessions.

This will **not** be done via email, for practical reasons due to cohort size.



Submission Rules

- ✓ Deliverable is an **Excel 2021 workbook** addressing all the tasks
- ✓ The model will be assessed on a University-spec. **PC-version** of Excel 2021 – Mac users take note!
- ✓ All answers should be presented in the correct cells on the **User** sheet
- ✓ The **User** sheet should also include all your ID numbers
- ✓ Only submit one model per group
- ✓ All answers **must be formula or function-based** i.e. **not** generated by **sorting** or **filtering** the data, **not** macro-based, **not** simple/static values, and **not** found manually by the user
- ✓ The exceptions are task 22, which can be static, and tasks 28 to 30 which *maybe* addressed by a macro but you **must include the VBA macros** in your submission to receive the marks
- ✓ Failure to comply with these rules will result in a **zero mark for that task**
- ✓ Only correct results will receive the marks. No partial marks will be awarded, unless stated otherwise



Penalties:

- | | |
|--|---------------------|
| ✗ The model should not exceed 8MB in size | (5 marks deducted) |
| ✗ The number of sheets in the workbook should not exceed 5 | (5 marks deducted) |
| ✗ The model should not have any hidden sheets | (5 marks deducted) |
| ✗ The model should be free from any circular references | (5 marks deducted) |
| ✗ All sheets should have a dedicated name , i.e. not simply Sheet1 | (5 marks deducted) |
| ✗ Using the incorrect dataset for your group | (10 marks deducted) |



Deadline for submission is:

16th January 2024

*** Important: Single-term students only ***

You must complete this coursework **individually**

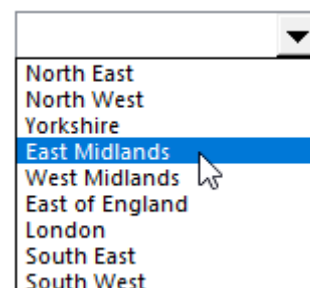
In addition to completing parts A, B, C and D, you must also complete the following task.



Part E – Creation of a New Datablock

[20 marks*]

Include a macro in your workbook that will create a new sheet containing just the data for a **chosen** Government Office Region (gor), selected by the user from a **drop-down box**



The new datablock should be constructed on a **newly-created** sheet, which should **not already exist** in the submitted workbook (hidden or otherwise).

The name of the new sheet should be the selected Government Office Region.

The macro should run off a button, located on the User sheet.



Ensure that the macro is bug-free and runs without crashing, can be run repeatedly without issues, and runs in a reasonable amount of time (<1 minute).

Deadline for single-term students is **the same as given on the previous page**

*Note: the marks for Parts B and C will be reduced to ensure the overall total remains out of 100 marks

Important: These data are copyrighted: no use must be made of this research example or the data contained therein without express permission. Furthermore, the data have been subject to small adjustments that maintain the general validity of any patterns observed but not necessarily the precise quantities. Dr Adam Hindle © 2023